

AI Internship Assignment: Agentic RAG System

Objective:

Build a lightweight AI agent capable of ingesting documents, creating embeddings, storing them in a vector database, and answering user queries using retrieved context.

Problem Statement:

Develop an AI-powered chatbot that:

- Ingest documents from multiple sources
- Processes and chunks the data
- Converts data into embeddings
- Stores embeddings in a vector database
- Retrieves relevant context based on user queries
- Generates accurate responses using an LLM

Input Data:

You can use a dataset containing:

- 10–20 documents (PDF, TXT, CSV)
- These may include SOPs, research papers, or product documentation

Core Requirements (Must Have):

- Document ingestion pipeline
- Text chunking and preprocessing
- Embedding generation
- Vector database implementation (FAISS/Chroma/Weaviate)
- Query interface (CLI or basic UI)
- Response generation using LLM

Bonus Features (Optional):

- Multi-source ingestion (Google Drive/API)
- Conversation memory
- Agentic behavior (tool usage)
- Web-based UI (Streamlit or simple frontend)

Technical Guidelines:

- Preferred Language: Python
- LLMs: OpenAI or open-source models
- Code must be modular and well-structured

Deliverables:

1. GitHub Repository:
 - Clean, well-documented code
 - README with architecture and workflow explanation
2. Demo Video (3–5 minutes):
 - Demonstrate system functionality
 - Show example queries and outputs
3. Write-up:
 - Architecture decisions
 - Limitations of your approach
 - Suggestions for scaling the system

Important Requirement:

The system should minimize hallucinations and provide answers strictly based on the ingested documents. Proper handling of unknown queries is expected.

Timeline:

Assignment Duration: 5 days

Submission includes all deliverables listed above

Submission Instructions:

Submit GitHub link, demo video link, and write-up document

Ensure all components are accessible and functional

We are looking for clarity of thought, strong fundamentals, and practical implementation skills.